Below is a **real-time, practical "When to use / When NOT to use" table** for major **AWS, Azure, and GCP cloud technology services**, mapped across real software development use cases.

✅ Cloud Services (AWS vs Azure vs GCP) — When to Use vs When NOT to Use (Real-time Development)

Legend
✅ Use it = Best-fit scenario
❌ Don't use it = Wrong fit / better alternative exists

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1) Compute (VM / Server)

Cloud Area	AWS Service	Azure Service	GCP Service	✅ When to Use (Real-time)	❌ When NOT to Use
Virtual Machines	EC2	Virtual Machines	Compute Engine	Hosting backend APIs, legacy apps, full control server, custom OS, VPN, custom middleware	If you want auto-scaling + no server management → use serverless / containers
Auto Scaling	Auto Scaling Group	VM Scale Sets	Managed Instance Groups	Auto-scaling app servers, traffic spikes	If workload is stable and fixed capacity
Bare Metal	Bare Metal (EC2 Dedicated Hosts)	Azure BareMetal	Bare Metal Solution	Ultra low-latency, licensed workloads	For normal apps → too expensive

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2) Serverless

Cloud Area	AWS	Azure	GCP	✅ When to Use	❌ When NOT to Use
Functions	Lambda	Azure Functions	Cloud Functions	Event-driven tasks: email triggers, image resize, cron jobs, microservices	For long-running heavy workloads (>15 mins) or persistent connections

Serverless Containers	Fargate	Container Apps	Cloud Run	Deploy APIs fast without managing servers, pay-per-request	If you need GPU / heavy ML compute (better use Kubernetes / VMs)
Workflow Orchestration	Step Functions	Logic Apps / Durable Functions	Workflows	Multi-step automation (order processing, approval flows)	For simple 1–2 step tasks

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3) Containers & Kubernetes

Cloud Area	AWS	Azure	GCP	✅ When to Use	❌ When NOT to Use
Managed Kubernetes	EKS	AKS	GKE	Enterprise microservices, scalable apps, DevOps pipelines, helm deployments	For small apps → Kubernetes complexity is unnecessary
Container Registry	ECR	ACR	Artifact Registry	Store Docker images securely for CI/CD	If you're not containerizing apps
Service Mesh	App Mesh	Open Service Mesh / Istio	Anthos Service Mesh	Secure microservices (mTLS), traffic routing	If you have only 2–3 services

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4) Storage (Files / Objects / Backup)

Cloud Area	AWS	Azure	GCP	✅ When to Use	❌ When NOT to Use
Object Storage	S3	Blob Storage	Cloud Storage	Store images/videos/files, logs, backup, static websites	Not for relational data querying (use DB)

File Storage	EFS	Azure Files	Filestore	Shared file storage (Linux/Windows)	For high-performance DB workloads
Archive	Glacier	Archive Storage	Archive Storage	Long-term cheap backup (years)	If you need fast access frequently

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5) Databases (SQL / NoSQL / Cache)

Cloud Area	AWS	Azure	GCP	✅ When to Use	❌ When NOT to Use
Managed SQL	RDS	Azure SQL Database	Cloud SQL	Most business apps, ERP/CRM, secure production DB	If huge scale globally → consider distributed DB
Cloud-native SQL	Aurora	Azure SQL MI	AlloyDB	High performance, autoscaling needs	For small apps—cost higher than basic SQL
NoSQL	DynamoDB	Cosmos DB	Firestore / Bigtable	High throughput: session store, carts, real-time events	If you need complex joins and SQL queries
Cache	ElastiCache (Redis)	Azure Cache for Redis	Memorystore	Speed boost: OTP/session storage, rate limiting	If data must be persistent forever
Data Warehouse	Redshift	Synapse	BigQuery	Analytics dashboards, reporting, BI	Not for transactional app backend

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6) Networking & API

Cloud Area	AWS	Azure	GCP	✅ When to Use	❌ When NOT to Use

Load Balancer	ALB/NLB/ ELB	Load Balancer	Cloud Load Balancing	Handle traffic, failover, secure routing	For low traffic single server
API Gateway	API Gateway	API Management	API Gateway	Manage APIs: auth, throttling, versioning	If direct internal service-to-service only
CDN	CloudFront	Azure CDN	Cloud CDN	Speed up global assets	For internal apps only
DNS	Route 53	Azure DNS	Cloud DNS	Production DNS routing	Not needed for local development

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7) Security / IAM

Cloud Area	AWS	Azure	GCP	✅ When to Use	❌ When NOT to Use
Identity & Access	IAM	Azure AD / Entra ID	IAM	Role-based permissions, secure access	Never give admin access to everyone
Secrets	Secrets Manager	Key Vault	Secret Manager	Store API keys, DB passwords securely	Don't store secrets in GitHub/env files
WAF	AWS WAF	Azure WAF	Cloud Armor	Protect websites from attacks	Not needed for private intranet apps

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8) Monitoring / Logging

Cloud Area	AWS	Azure	GCP	✅ When to Use	❌ When NOT to Use
Monitoring	CloudWatch	Azure Monitor	Cloud Monitoring	Production system health monitoring	Not optional in production
Central Logs	CloudWatch Logs	Log Analytics	Cloud Logging	Debug errors, audit access, detect attacks	Don't rely only on application console logs

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9) Messaging / Streaming

Cloud Area	AWS	Azure	GCP	✅ When to Use	❌ When NOT to Use
Queue	SQS	Storage Queue	Pub/Sub	async processing: email, notifications, order processing	If strict ordering is required (use Kafka or FIFO queue)
Streaming	Kinesis	Event Hubs	Pub/Sub	real-time logs, IoT events, clickstream	For simple tasks → too complex

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10) DevOps / CI-CD

Cloud Area	AWS	Azure	GCP	✅ When to Use	❌ When NOT to Use
CI/CD	CodePipeline	Azure DevOps	Cloud Build	Automated build/test/deploy	If team already uses GitHub Actions/Jenkins
IaC	CloudFormation	ARM/Bicep	Deployment Manager/ Terraform	Infrastructure provisioning repeatably	Never create production infra manually

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✅ Quick Decision Guide (Real-world)

If your company/project is…	Best Cloud
Microsoft-based (Windows, AD, Office 365, .NET)	✅ Azure
Huge variety of services + mature ecosystem	✅ AWS
Data analytics + ML + BigQuery heavy use	✅ GCP
Hybrid on-prem + cloud + enterprise governance	✅ Azure / GCP Anthos
Startups needing fastest production scaling	✅ AWS / GCP

If you want, I can also provide:
✅ **AWS vs Azure vs GCP service-to-service mapping table (100+ services)**
✅ **Architecture diagrams for real apps** (E-commerce, WhatsApp chat, ERP, AI app)
✅ **Which services to use for your Django / Java / Kubernetes projects** (based on your earlier work)

Absolutely ✅— here is a **100+ services AWS vs Azure vs GCP mapping table** (service-to-service equivalence / closest match), grouped by real cloud categories.

 Note: Not every service has a *perfect* 1-to-1 match across all 3 clouds. In such cases I wrote **Closest equivalent / Alternative**.

Sources used for mapping: Google Cloud official comparison + Microsoft AWS-to-Azure mappings. (Google Cloud Documentation)

✅ AWS vs Azure vs GCP — Service-to-Service Mapping Table (120+)

1) Core / Accounts / Management

Category	AWS	Azure	GCP
Account mgmt	AWS Organizations	Management Groups	Resource Manager + Organizations
Billing	AWS Cost Explorer / Billing Console	Cost Management + Billing	Cloud Billing
Policy / Governance	AWS IAM + SCP	Azure Policy	Organization Policy Service
Resource templates (IaC)	CloudFormation	ARM Templates / Bicep	Deployment Manager *(legacy)* / Terraform
CLI	AWS CLI	Azure CLI	gcloud CLI
Console UI	AWS Console	Azure Portal	Google Cloud Console
Resource tagging	Tags	Tags	Labels
Audit API logs	CloudTrail	Activity Log	Cloud Audit Logs
Config tracking	AWS Config	Azure Policy / Resource Graph	Cloud Asset Inventory
Cloud advisor	Trusted Advisor	Advisor	Recommender

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2) Compute (VM / Scaling / OS)

Category	AWS	Azure	GCP
Virtual Machines	EC2	Virtual Machines	Compute Engine
Auto Scaling	EC2 Auto Scaling	VM Scale Sets	Managed Instance Groups
Spot instances	EC2 Spot	Spot VMs	Spot VMs
Dedicated hardware	Dedicated Hosts	Dedicated Host	Sole-Tenant Nodes
VM images	AMI	Azure Images	Images
VM marketplace	AWS Marketplace	Azure Marketplace	Cloud Marketplace

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3) Containers / Kubernetes

Category	AWS	Azure	GCP
Kubernetes	EKS	AKS	GKE
Container runtime service	ECS	Container Instances	Cloud Run / GKE Autopilot (closest)
Serverless containers	Fargate	Container Apps	Cloud Run
Container registry	ECR	ACR	Artifact Registry
Service mesh	App Mesh	Istio / Open Service Mesh	Anthos Service Mesh
Multi-cluster mgmt	EKS Anywhere	Azure Arc	Anthos

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4) Serverless / Functions

Category	AWS	Azure	GCP
Functions	Lambda	Azure Functions	Cloud Functions
Event routing	EventBridge	Event Grid	Eventarc
Workflow orchestration	Step Functions	Logic Apps	Workflows

API for serverless	API Gateway	API Management	API Gateway

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5) Storage (Object / File / Block / Archive)

Category	AWS	Azure	GCP
Object storage	S3	Blob Storage	Cloud Storage
Block storage	EBS	Managed Disks	Persistent Disk
File storage	EFS	Azure Files	Filestore
Archive	S3 Glacier	Blob Archive tier	Archive Storage
Storage transfer	DataSync	Storage Migration Service	Storage Transfer Service
Backup	AWS Backup	Azure Backup	Backup and DR Service

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6) Networking (VPC / Routing / LB / DNS / CDN)

Category	AWS	Azure	GCP
Virtual network	VPC	VNet	VPC
Subnet	Subnet	Subnet	Subnet
Firewall rules	Security Groups / NACL	NSG	VPC Firewall Rules
NAT	NAT Gateway	NAT Gateway	Cloud NAT
Private connectivity	PrivateLink	Private Link	Private Service Connect
Load balancer	ELB/ALB/NLB	Azure Load Balancer / App Gateway	Cloud Load Balancing
DNS	Route 53	Azure DNS	Cloud DNS
CDN	CloudFront	Azure CDN	Cloud CDN
DDoS	AWS Shield	DDoS Protection	Cloud Armor

VPN	VPN Gateway	VPN Gateway	Cloud VPN
Dedicated link	Direct Connect	ExpressRoute	Cloud Interconnect
Network monitoring	VPC Flow Logs	NSG Flow Logs	VPC Flow Logs

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7) Identity / Security / Keys / Secrets

Category	AWS	Azure	GCP
Identity & roles	IAM	Entra ID + RBAC	Cloud IAM
MFA / SSO	IAM Identity Center	Entra ID	Cloud Identity
Key management	KMS	Key Vault	Cloud KMS
Secret mgmt	Secrets Manager	Key Vault Secrets	Secret Manager
Hardware HSM	CloudHSM	Dedicated HSM	Cloud HSM
WAF	AWS WAF	Azure WAF	Cloud Armor
Security posture	Security Hub	Defender for Cloud	Security Command Center
Encryption policy	KMS + Policies	Policy + Key Vault	CMEK + Policies

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8) Databases — Relational (SQL)

Category	AWS	Azure	GCP
Managed MySQL/Postgres	RDS	Azure Database for MySQL/PostgreSQL	Cloud SQL
Enterprise SQL Server	RDS SQL Server	Azure SQL Database	Cloud SQL SQL Server
Cloud-native SQL	Aurora	Azure SQL Managed Instance	AlloyDB (Postgres)
Globally distributed SQL	Aurora Global / RDS read replicas	Azure SQL Hyperscale	Spanner

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9) Databases — NoSQL / Key-Value / Document

Category	AWS	Azure	GCP
Key-value NoSQL	DynamoDB	Cosmos DB	Bigtable *(closest)*
Document DB	DocumentDB	Cosmos DB (Mongo API)	Firestore
Managed Mongo	DocumentDB	Cosmos DB Mongo API	Mongo Atlas (Marketplace)
In-memory cache	ElastiCache (Redis/Memcached)	Azure Cache for Redis	Memorystore
Graph DB	Neptune	Cosmos DB (Gremlin)	Neptune alternative *(no native)*

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10) Data Analytics / Warehousing / ETL

Category	AWS	Azure	GCP
Data warehouse	Redshift	Synapse Analytics	BigQuery
Spark / Big Data	EMR	HDInsight / Databricks	Dataproc
Stream processing	Kinesis Data Analytics	Stream Analytics	Dataflow
Batch processing	AWS Batch	Batch	Batch
ETL	Glue	Data Factory	Data Fusion
Data catalog	Glue Data Catalog	Purview	Data Catalog
BI dashboards	QuickSight	Power BI	Looker
Data lake	Lake Formation	ADLS Gen2	Cloud Storage + BigLake

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11) Messaging / Events / Integration

Category	AWS	Azure	GCP
Queue	SQS	Storage Queues / Service Bus	Pub/Sub
Enterprise messaging	Amazon MQ	Service Bus	Pub/Sub + partner
Pub/Sub events	SNS	Event Grid	Pub/Sub
Kafka	MSK	Event Hubs (Kafka API)	Pub/Sub + Confluent
iPaaS integration	AppFlow	Logic Apps	Application Integration

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12) Monitoring / Logging / Observability

Category	AWS	Azure	GCP
Monitoring	CloudWatch	Azure Monitor	Cloud Monitoring
Logging	CloudWatch Logs	Log Analytics	Cloud Logging
Tracing	X-Ray	Application Insights	Cloud Trace
Error reporting	CloudWatch alarms	Alerts	Error Reporting
Metrics + alerts	CloudWatch	Alerts	Alerting
APM	X-Ray + partners	App Insights	Operations Suite

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13) DevOps / CI-CD / Artifacts

Category	AWS	Azure	GCP
Git repo	CodeCommit	Azure Repos	Cloud Source Repositories
CI build	CodeBuild	Pipelines	Cloud Build
CD deploy	CodeDeploy	Pipelines / Release	Cloud Deploy
Artifacts	CodeArtifact	Azure Artifacts	Artifact Registry

Dev environments	Cloud9	Dev Box	Cloud Workstations

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14) AI / ML / GenAI

Category	AWS	Azure	GCP
ML platform	SageMaker	Azure Machine Learning	Vertex AI
GenAI platform	Amazon Bedrock	Azure OpenAI	Vertex AI (Gemini)
Vision	Rekognition	Azure AI Vision	Vision AI
OCR / docs	Textract	Document Intelligence	Document AI
Speech-to-text	Transcribe	Speech Services	Speech-to-Text
Text-to-speech	Polly	Speech Services	Text-to-Speech
Translation	Translate	Translator	Translation API
Chatbots	Lex	Bot Service	Dialogflow
Data labeling	Ground Truth	Data labeling	Vertex AI Data Labeling

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15) Search / Discovery

Category	AWS	Azure	GCP
Search service	OpenSearch Service	Azure AI Search	Vertex AI Search / Enterprise Search
Log analytics search	OpenSearch	Log Analytics	Logging + BigQuery

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16) Content Delivery / Media Services

Category	AWS	Azure	GCP
Media transcoding	Elemental MediaConvert	Media Services	Transcoder API

Live streaming	MediaLive	Media Services live	Live Stream API (via partners)

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17) IoT / Edge

Category	AWS	Azure	GCP
IoT device hub	IoT Core	IoT Hub	IoT Core *(retired)* / partners
Edge runtime	Greengrass	IoT Edge	Anthos at the edge

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18) Mobile / Notifications

Category	AWS	Azure	GCP
Push notifications	SNS Mobile Push	Notification Hubs	Firebase Cloud Messaging
App backend	Amplify	App Service + Mobile Apps	Firebase

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19) API / Application Layer

Category	AWS	Azure	GCP
API gateway	API Gateway	API Management	API Gateway
App hosting (PaaS)	Elastic Beanstalk	App Service	App Engine
App config	AppConfig	App Configuration	Runtime Config *(legacy)* / Secret Manager

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20) Migration / Hybrid / DR

Category	AWS	Azure	GCP
VM migration	Application Migration Service	Azure Migrate	Migrate to Virtual Machines
DB migration	DMS	Database Migration Service	Database Migration Service

Backup vault	AWS Backup	Recovery Services Vault	Backup and DR
Hybrid ops	Outposts	Azure Arc	Anthos